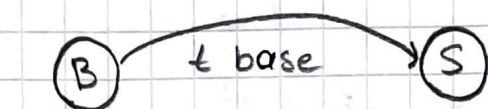


NUMBER SYSTEMS



Element
Set

$$b \in B$$

Symbol
Set

$$a_i \in S$$

$$a_i \in \{0, 1, 2, \dots, t-1\}$$

Fixed Point Number System

Floating Point Number System

$$1) \quad b = S \cdot (a_n t^n + \dots + a_1 t^1 + a_0 + a_{-1} t^{-1} + \dots + a_{-l} t^{-l})$$

$$2) \quad b = S \cdot \sum_{i=-1}^n a_i \cdot t^i$$

$$3) \quad b = S \cdot (a_n \cdot a_{n-1} \dots a_1 a_0 \cdot a_{-1} \dots a_{-l})$$

$$t=2 \quad \text{Binary Number System} \quad a_i \in \{0, 1\}$$

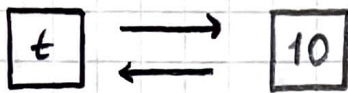
$$t=8 \quad \text{Octal Number System} \quad a_i \in \{0, 1, \dots, 7\}$$

$$t=10 \quad \text{Decimal Number System} \quad a_i \in \{0, 1, \dots, 9\}$$

$$t=16 \quad \text{Hexadecimal Number System} \quad a_i \in \{0, 1, \dots, 9, A, B, C, \dots, F\}$$

$$b_k = S \cdot \underbrace{a_n a_{n-1} \dots a_0}_{\text{integer}} \cdot \underbrace{a_{-1} \dots a_{-l}}_{\text{fractional}}$$

↓ sign
MSB ↓
LSB ↓
MSB ↑
LSB ↑



2, 8, 16



$$\begin{aligned} 8 &= 2^3 \\ 16 &= 2^4 \end{aligned}$$



$t \rightarrow 10$

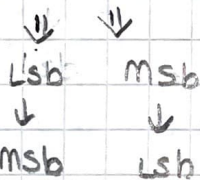
$$b = (11010, 10110)_2 \rightarrow ?$$

$$b = 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^1 + 1 \cdot 2^{-1} + 1 \cdot 2^{-3} + 1 \cdot 2^{-4}$$

$$b = (26,6875)_{10}$$

$$\Rightarrow 10 \rightarrow t(2, 8, 16)$$

$$b = N.M$$



ÜBN $(353)_{10} = (\dots)_2$

353	a_i	
176	1	$a_0 \rightarrow \text{LSB}$
88	0	
44	0	
22	0	
11	0	
5	1	
2	1	
1	0	
0	1	$a_8 \rightarrow \text{MSB}$

FRACTIONAL CONVERSION

$$r^{-p} = t^{-q}$$

$$q = p \cdot \frac{\ln r}{\ln t} \Rightarrow q = p \cdot \frac{\ln 10}{\ln 2} \approx 3.32 p$$

$$m = (0.65625)_{10} = (\dots)_2$$

$$q = (3.32) \cdot 5 = 16.6 \approx 17$$

q_i	p_i	t
-	0,65625	2
1	0,31250	
0	0,62500	
1	0,25000	
0	0,50000	
1	0,00000	

(0,10101)

$$\cdot \quad 2 \longleftrightarrow \begin{array}{|c|} \hline 8 \\ \hline 16 \\ \hline \end{array} \quad \begin{array}{l} 2^3 = 8 \\ 2^4 = 16 \end{array}$$

ÖRN $(100111010)_2 = (\dots)_8 = (\dots)_?$

$$(100111010)_2 = (\dots)_{16} = (13A)_{16}$$

$$(BA92)_{16} = (\dots)_2$$

$$(BA92)_{16}$$

$$\begin{array}{cccc} \downarrow & \downarrow & & \\ (1011 & 1010 & 1001 & 0010)_2 \\ B & A & 9 & 2 \end{array}$$

$$\begin{array}{r} 16 \quad 5848 \\ 2+8+16+32 \\ +256 \\ \hline 314 \quad 8 \\ -24 \quad 39 \quad 8 \\ \hline 74 \quad 32 \quad 4 \quad 8 \\ -72 \quad 2 \quad 4 \quad 8 \\ \hline 2 \quad 2 \quad 4 \quad 8 \end{array}$$

$$\begin{array}{r} 8. \\ 64 \\ \times 8 \\ \hline 2 \end{array}$$

THE COMPLEMENTATION OF THE NUMBERS

→ Complementation to $t-1$ (Radix minus one notation)

→ Complementation to t (Radix notation)

COMPLEMENTATION to $t-1$

$$T = t^n - t^{-m} - N$$

$$N = 12,79$$

$\underbrace{\quad\quad}_n \quad \underbrace{\quad\quad}_m$

$$T = 10^2 - 10^{-2} - 12,79 = 87,20$$

$$\begin{array}{r} N \quad 12,79 \\ + T \quad 87,20 \\ \hline 99,99 \end{array}$$

$$N = (101)_2 \Rightarrow T = 010$$

$$(654)_{10} \Rightarrow T = (345)_{10} \rightarrow (999 - 345)$$

COMPLEMENTATION to t

$$T_t = t^n - N = T_{t-1} + 1(\text{LSB})$$

$$N = 12,79, \quad T_t = 10^4 - 1279 = 87,21$$

$$T_t = T_{t-1} + 1(\text{LSB}) = 87,20 + 1 = 87,21$$

$$N = (101)_2 \Rightarrow \begin{array}{l} T_1 = 010 \\ T_2 = 011 \end{array}$$

NEGATIVE NUMBER

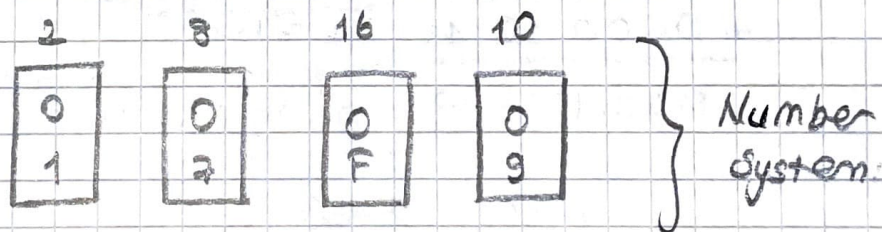
Sign and magnitude

Diminished - Radix Notation (Radix minus One)

Radix Notation

	<u>s + m</u>	<u>t - 1</u> ^{15'e} tamamlıyor	<u>t</u> ^{16'ya} tamamlıyor
+5	0 0101 magnitude	0 0101	0 0101
-5	1 0101 sign	1 1010	1 1011

	<u>s + m</u>	<u>t - 1</u>	<u>t</u>
+9	0 1001	0 1001	0 1001
-9	1 1001	1 0110	1 0 111



FIXED POINT ARITHMETIC

→ Radix Notation (t - base)

→ Radix minus One (t - 1 base)

case 1 A > 0 B > 0

case 2 A > 0 B < 0 (Vice versa)

case 3 A < 0 B < 0

Radix Notation $\rightarrow$ 2'ye t kleyen

$\rightarrow$ case $A > 0$ $B > 0$

$$\begin{array}{r} A \quad 0 \quad a_{n-2} \cdot a_{n-1} \dots a_1 a_0, a_{-1} \dots a_{-k} \\ + B \quad 0 \quad b_{n-2} \cdot b_{n-3} \dots b_1 b_0, b_{-1} \dots b_{-k} \\ \hline \end{array}$$

$$S_s \quad S_{n-2} \cdot S_{n-3} \dots S_1 S_0, S_{-1} \dots S_{-k}$$

$$S_s = 0 \quad A + B < t^{n-1}$$

$$S_s \neq 0 \quad A + B > t^{n-1} \quad \text{over flow}$$

 RN

$$\begin{array}{r} 13 \\ + 11 \\ \hline 24 \end{array}$$

$$\begin{array}{r} 01101 \\ + 01011 \\ \hline 11000 \end{array}$$

over flow

$$\begin{array}{r} 00001101 \\ + 00001011 \\ \hline 00011000 \end{array}$$

8 bit

yer ayrilmas  lsaydı

$\rightarrow$ case $A < 0$, $B < 0$

 RN

$$\begin{array}{r} -A \quad (t-1) \cdot a_{n-2} \dots a_1 a_0, a_{-1} \dots a_{-t} \\ + -B \quad (t-1) \cdot b_{n-2} \dots b_1 b_0, b_{-1} \dots b_{-k} \\ \hline - (A+B) \end{array}$$

$$C_s \quad S_s \quad S_{n-2} \dots S_1 S_0, S_{-1} \dots S_{-k}$$

Carry

ÖRN

$$\begin{array}{r} t^n - A \\ + t^n - B \\ \hline \end{array}$$

$$2t^n - (A+B)$$

$$t^n + [t^n - (A+B)]$$

$$\begin{array}{r} A - 4 \\ + B - 7 \\ \hline \end{array}$$

$$-11$$

ÖRN

st m

(15)

t - 1

(16)

t

$$-4 \quad 10100$$

$$11011$$

$$11100$$

ÖRN

$$\begin{array}{r} 11100 \\ + 11001 \\ \hline 1110101 \end{array}$$

ignore

→ case A > 0, B < 0

$$\begin{array}{r} A \\ + -B \\ \hline \end{array}$$

$$\begin{array}{r} A \\ + t^n - B \\ \hline t^n - (B - A) \end{array}$$

ÖRN

$$\begin{array}{r} 15 \\ + -6 \\ \hline 9 \end{array}$$

$$\begin{array}{r} 01111 \\ + 11010 \\ \hline 101001 \end{array}$$

ignore

ÖRN

$$\begin{array}{r} -15 \\ + 6 \\ \hline -9 \end{array}$$

$$\begin{array}{r} 10001 \\ + 00110 \\ \hline 10111 \end{array}$$

FIXED POINT ARITHMETIC

Diminished Radix Arithmetic Base

→ $A > 0 \quad B > 0$

$$\begin{array}{r} +A \\ +B \\ \hline + (A+B) \end{array}$$

$$\begin{array}{r} 4 \\ + 11 \\ \hline + 15 \end{array}$$

ilk sütun
işaret biti

kalan dördü
binary
karsılığı

$$\begin{array}{r} 00100 \\ + 01011 \\ \hline 01111 \end{array}$$

ÖRN

$$\begin{array}{r} +6 \\ + 11 \\ \hline + 17 \end{array}$$

$$\begin{array}{r} 00100 \\ + 01011 \\ \hline 10001 \\ \text{overflow} \end{array}$$

→ $A < 0 \quad B < 0$

$$\begin{array}{r} -A \\ -B \\ \hline - (A+B) \end{array}$$

$$\begin{array}{r} t^n - t^{-k} - A \\ t^n - t^{-k} - B \\ \hline 2t^n - 2t^{-k} - (A+B) \end{array}$$

↓

End Around
carry

olması gereken

$$t^n - t^{-k} - (A+B) \quad t^n - t^{-k} + [t^n - t^{-k} - (A+B)]$$

msb lsb fazlalık

ÖRN

$$\begin{array}{r} -3 \\ + -11 \\ \hline -14 \end{array}$$

$$\begin{array}{r} 11100 \\ + 10100 \\ \hline 110000 \\ + 1 \\ \hline 10001 \end{array}$$

$$+3 \rightarrow 0011$$

2'ye
2'te

$$-3 \rightarrow 1100$$

~örneğin devamı

$$11 \rightarrow 1011$$

↓ 2'thi

$$-11 \rightarrow 10100$$

işaret biti

$$14 \rightarrow 1110$$

↓ 2'thi

$$-14 \rightarrow 10001$$

işaret biti

→ $A > 0$ $B < 0$ (vice varsa)

$$\begin{array}{r} A \\ + -B \\ \hline \end{array}$$

$$\begin{array}{r} A \\ + t^n - t^k - B \\ \hline t^n - t^k - (B - A) \end{array}$$

ÖRN

$$\begin{array}{r} -3 \\ + 11 \\ \hline 8 \end{array}$$

$$\begin{array}{r} 11100 \\ + 01011 \\ \hline 100111 \end{array}$$

End
around
carry

$$\begin{array}{r} 100111 \\ + 1 \\ \hline 01000 \end{array}$$

karşılığı alırken
yapıyoruz (2'ye kullandık)

$$8 \rightarrow 01000$$

ÖRN

$$\begin{array}{r} -11 \\ + 3 \\ \hline -8 \end{array}$$

$$\begin{array}{r} 10100 \\ + 00011 \\ \hline 10111 \end{array}$$

$$8 \rightarrow 1000$$

$-8 \rightarrow 10111 \Rightarrow$ 2'thini al başına bit işaretini koy.

$$\begin{array}{r} - \rightarrow 1 \\ + \rightarrow 0 \end{array}$$

End around carry yok

$$5 \text{ bit} \rightarrow -16 \sim +15$$

$$4 \text{ bit} \rightarrow -8 \sim +7$$

$$\rightarrow \underbrace{1000 \dots 0000b}_{-2^{8-1}} \rightarrow -128$$

$$+127$$

$$-2^{n-1} \sim +2^{n-1} - 1$$

ÖRN

$$110101$$

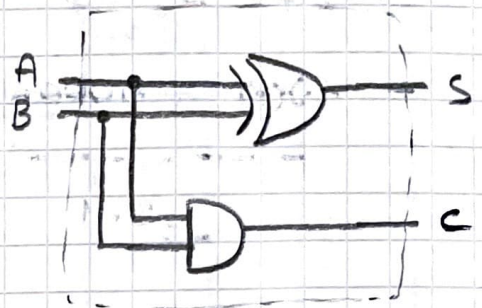
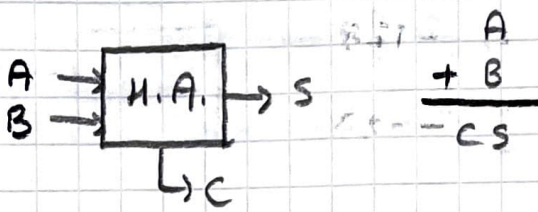
$$S + m \rightarrow 110101 \rightarrow -21$$

- 1's Complement $\rightarrow \underbrace{110101}_{\text{tersini al}} \rightarrow 01010 \rightarrow -10$
1 fazlası ↑

- 2's Complement $\rightarrow 110101 \rightarrow 01010 \rightarrow -11$

$$\begin{array}{r} + \quad 1 \\ \hline 01011 \\ \hline 11 \end{array}$$

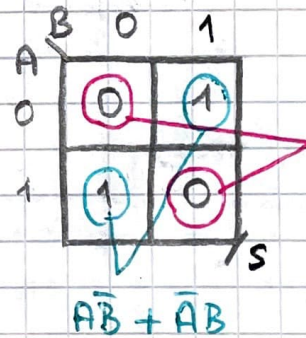
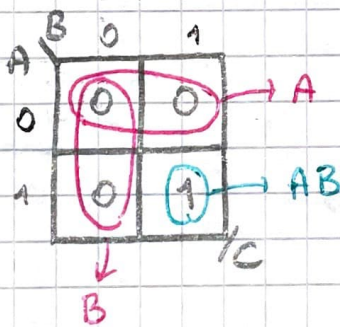
HALF ADDER



Dec	A	B	C (ekle (carry))	S (toplam (sum))
0	0	0	0	0
1	0	1	0	1
2	1	0	0	1
3	1	1	1	0

a) Yalnızca NAND

b) Yalnızca NOR kapıları kullanarak gerçekleştiriniz.

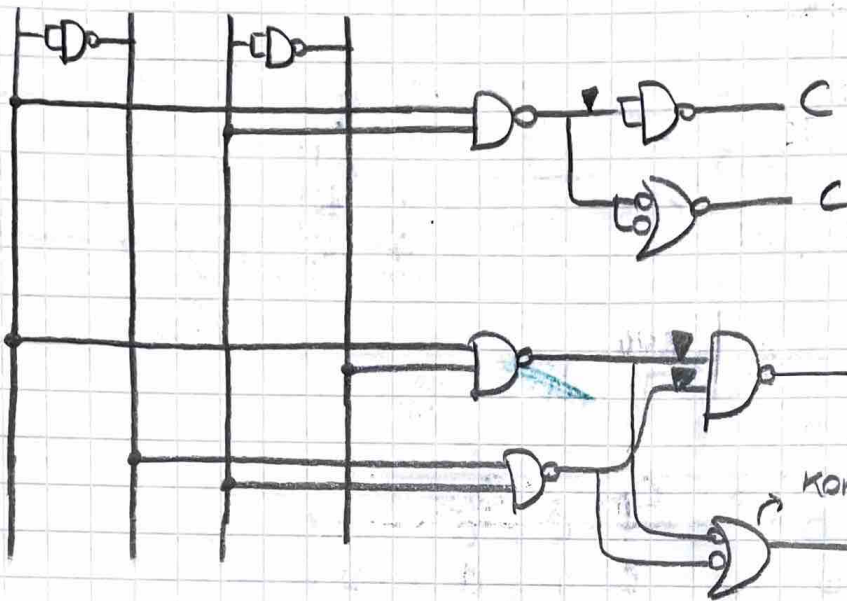


$$a) C = AB \Rightarrow C = \overline{\overline{AB}}$$

$$S = \overline{A}B + A\overline{B} \Rightarrow S = \overline{\overline{\overline{A}B} \cdot \overline{A\overline{B}}}$$

$$S = \overline{\overline{A}B} \cdot \overline{A\overline{B}}$$

A $\bar{A}$ B $\bar{B}$

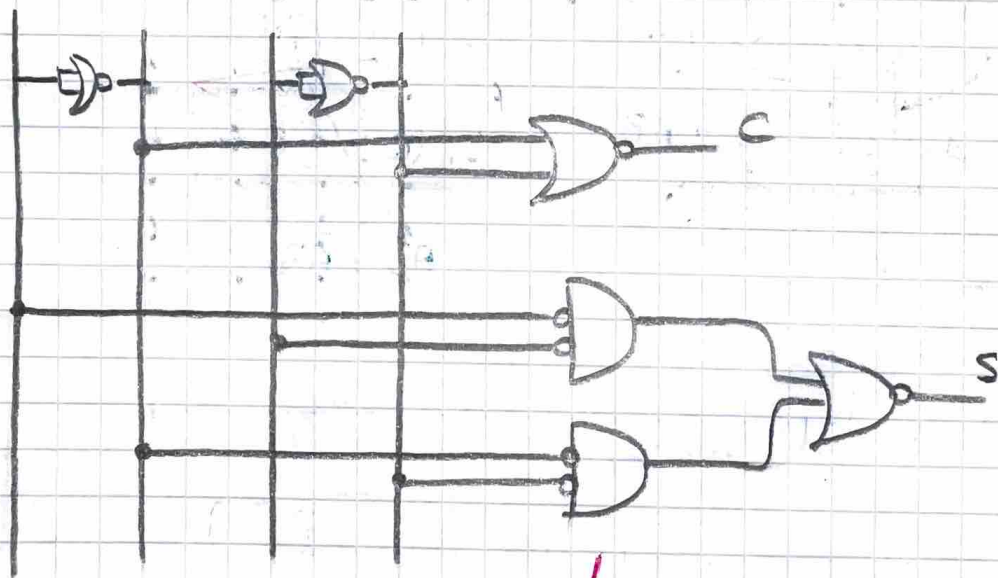


Kongugate (logic uyumsuzluq)

$$b) C = AB \Rightarrow C = \overline{\overline{AB}} = \overline{\bar{A} + \bar{B}}$$

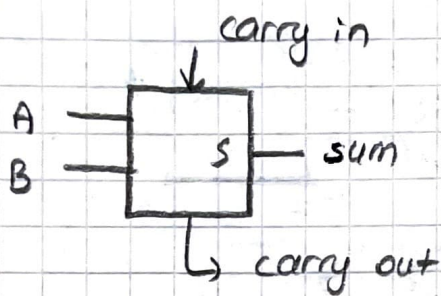
$$S = (A+B) \cdot (\bar{A} + \bar{B}) = \overline{(A+B) \cdot (\bar{A} + \bar{B})} = \overline{(A+B)} + \overline{(\bar{A} + \bar{B})}$$

A $\bar{A}$ B $\bar{B}$



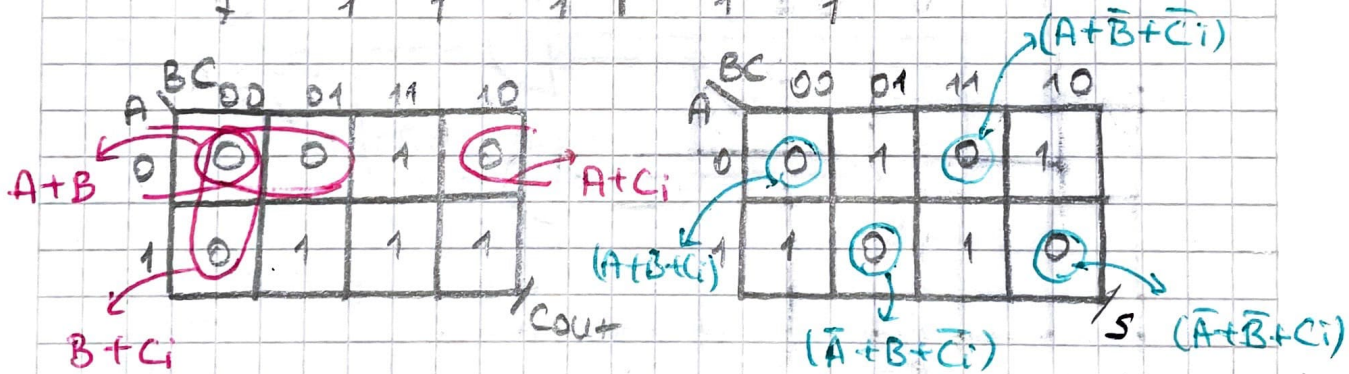
NOR
kongugate

FULL ADDER



$$\begin{array}{r} \text{Cin} \\ A \\ + B \\ \hline S \rightarrow \text{Cout} \end{array}$$

Dec	A	B	Cin	Cout	S
0	0	0	0	0	0
1	0	0	1	0	1
2	0	1	0	0	1
3	0	1	1	1	0
4	1	0	0	0	1
5	1	0	1	1	0
6	1	1	0	1	0
7	1	1	1	1	1



$$C_{out} = (A+B) \cdot (A+Ci) \cdot (B+Ci)$$

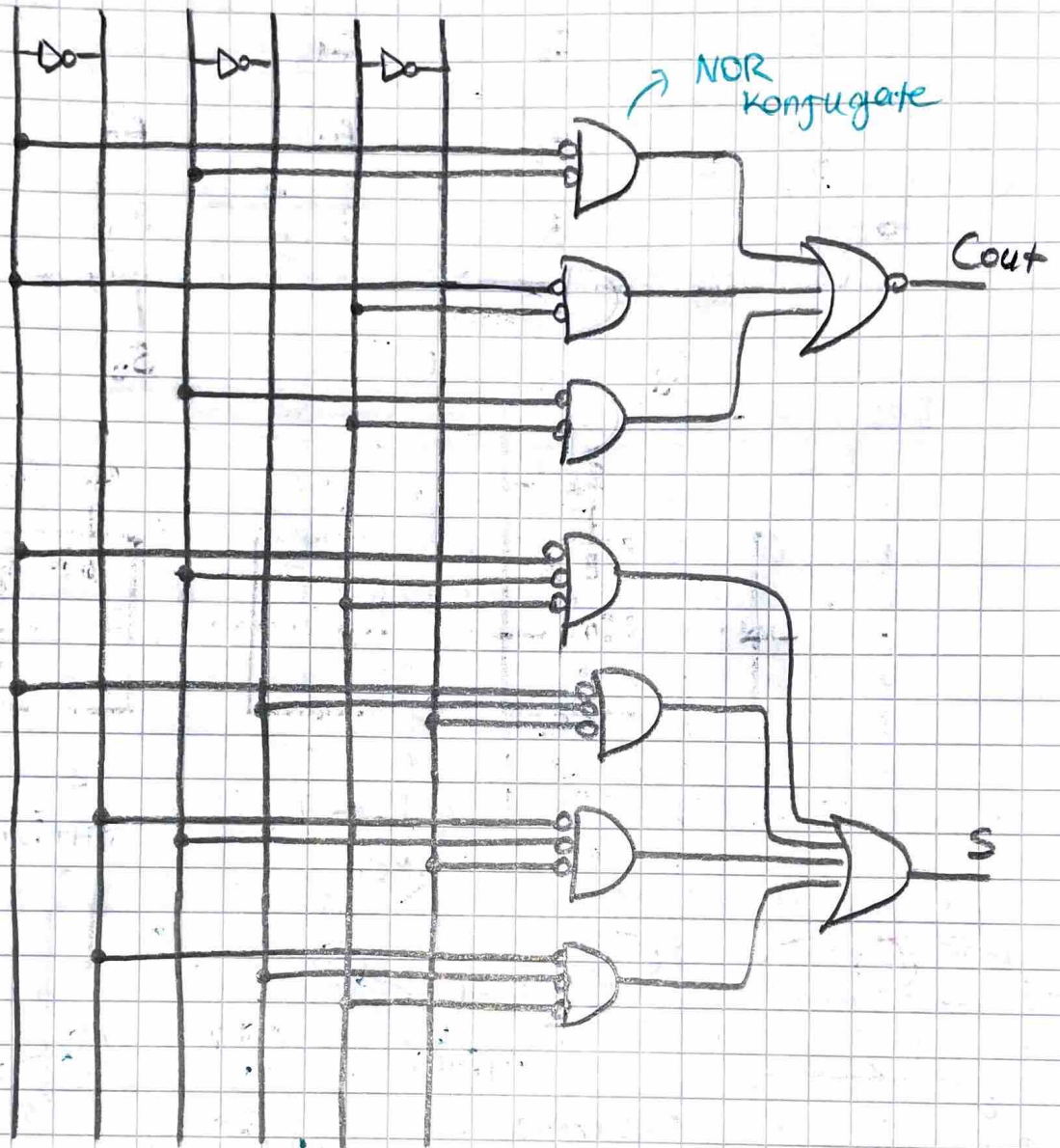
$$C_{out} = \overline{(A+B)} + \overline{(A+Ci)} + \overline{(B+Ci)}$$

$$S = (A+B+Ci) \cdot (A+\bar{B}+\bar{C}i) \cdot (\bar{A}+B+\bar{C}i) \cdot (\bar{A}+\bar{B}+Ci)$$

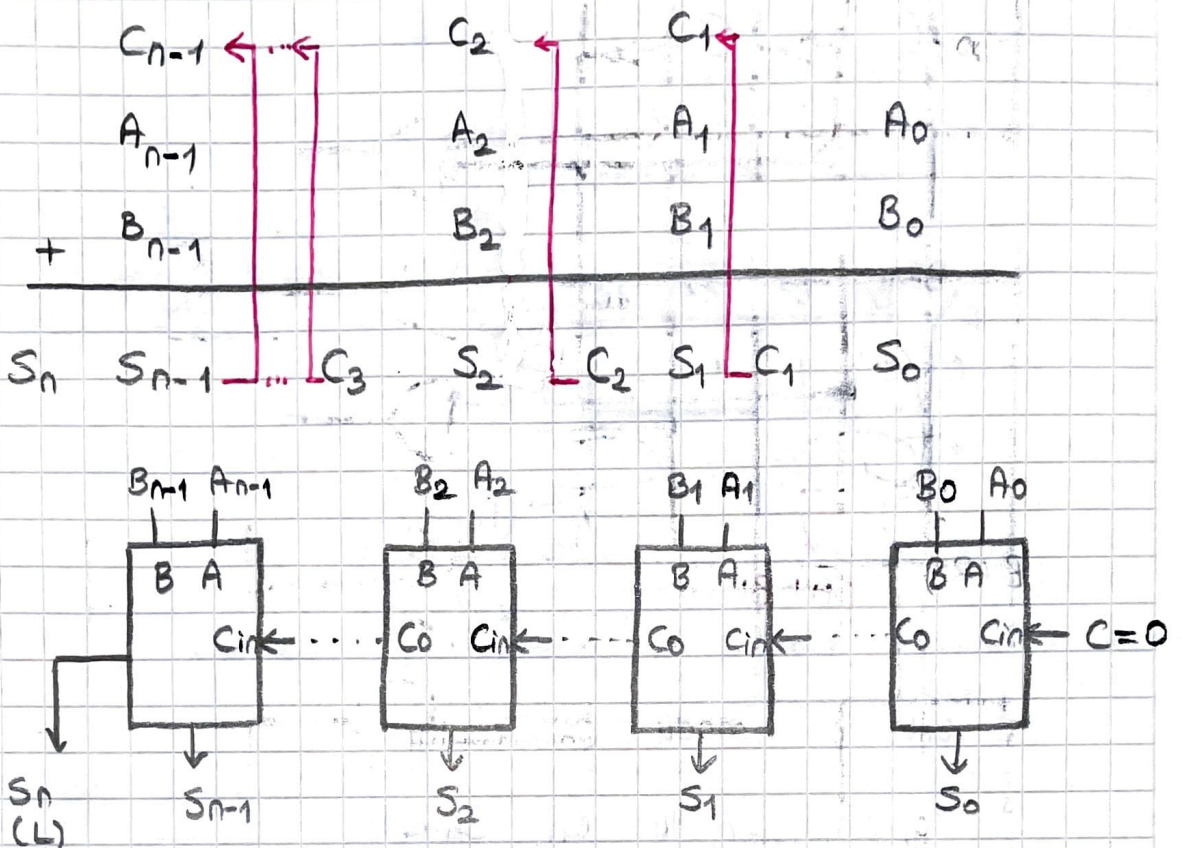
$$S = \overline{(A+B+Ci)} + \overline{(A+\bar{B}+\bar{C}i)} + \overline{(\bar{A}+B+\bar{C}i)} + \overline{(\bar{A}+\bar{B}+Ci)}$$

Devrenin Giziimi :

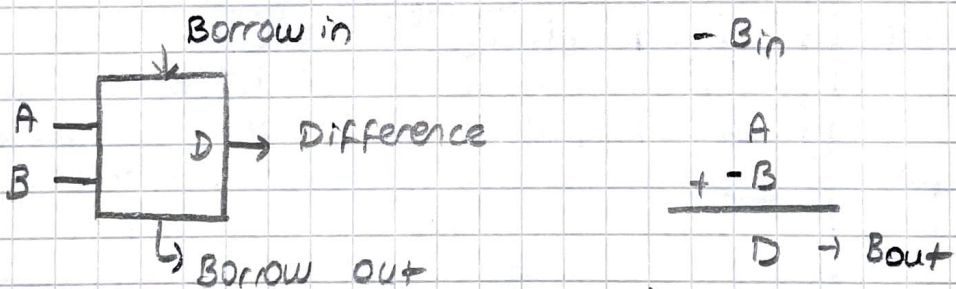
A $\bar{A}$ B $\bar{B}$ C $\bar{C}$



n BIT PARALEL ADDER



FULL SUBTRACTOR



Dec	A	B	Bin	D	Bo
0	0	0	0	0	0
1	0	0	1	1	1
2	0	1	0	1	1
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	0
6	1	1	0	0	0
7	1	1	1	1	1

→ 2'den 2 çıktı $D=0$
Bora bu yüzden $B_o=1$

→ A'don $(B + Bin)$ çıkar → $A - (B + Bin)$

	BB _i	00	01	11	10
A		0	1	0	1
		1	0	1	0

	$B\bar{B}_i$	00	01	11	10
A					
0	0	1	1	1	
1	0	0	1	0	

$(B + B_i)$ (blue arrow from 0,0 to 1,0)
 $(\bar{A} + B)$ (red arrow from 0,0 to 1,1)
 B_0 (blue arrow from 1,1 to 1,0)
 $(\bar{A} + B_i)$ (blue arrow from 1,0 to 1,1)

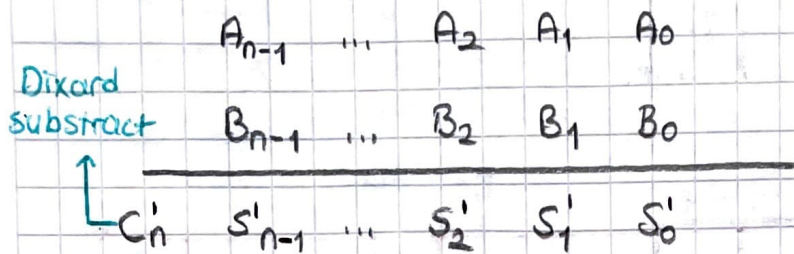
→ Çıkarma modu C_n toplanır

→ Toplama modu C_n S_n 'dir.

COMPLEMENT ADDER

ADDER SUBTRACTOR

Burayı tekrar kontrol et,



subtract
K' complement

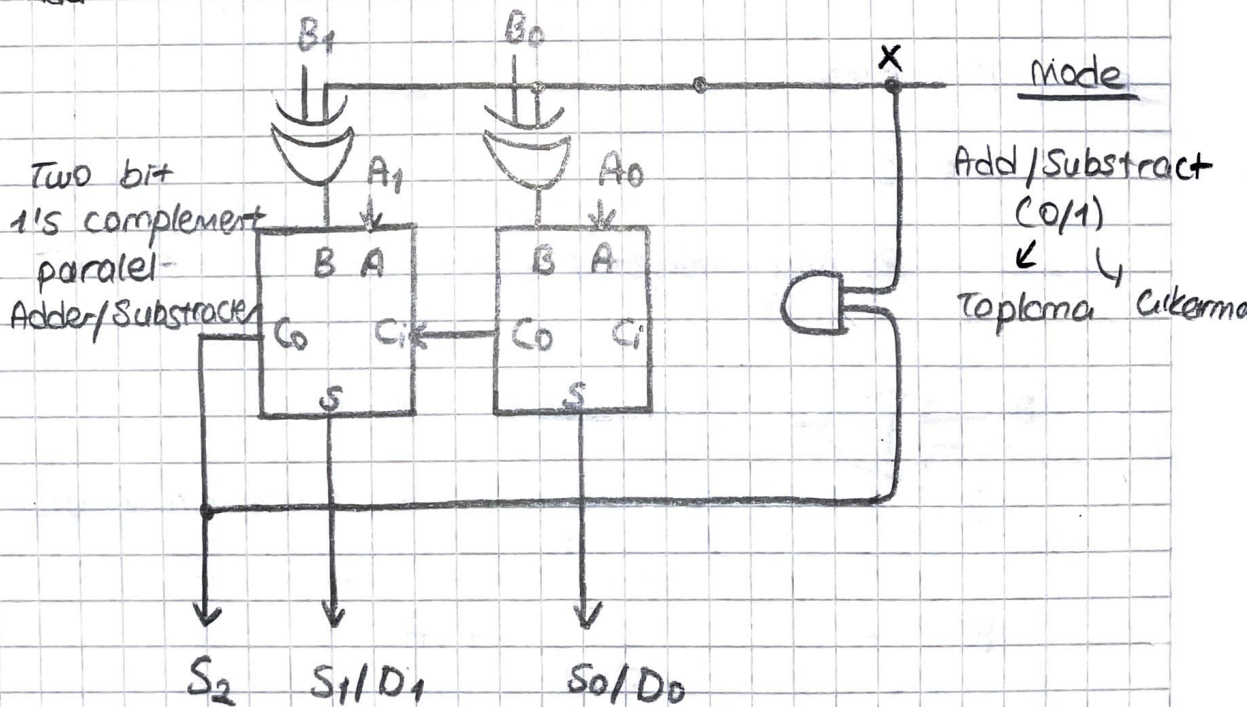
1's complement plus C'_n

first sum çıkartma modu

C'_n - biti toplama yapınca S_n olur. Carry end Around and A add

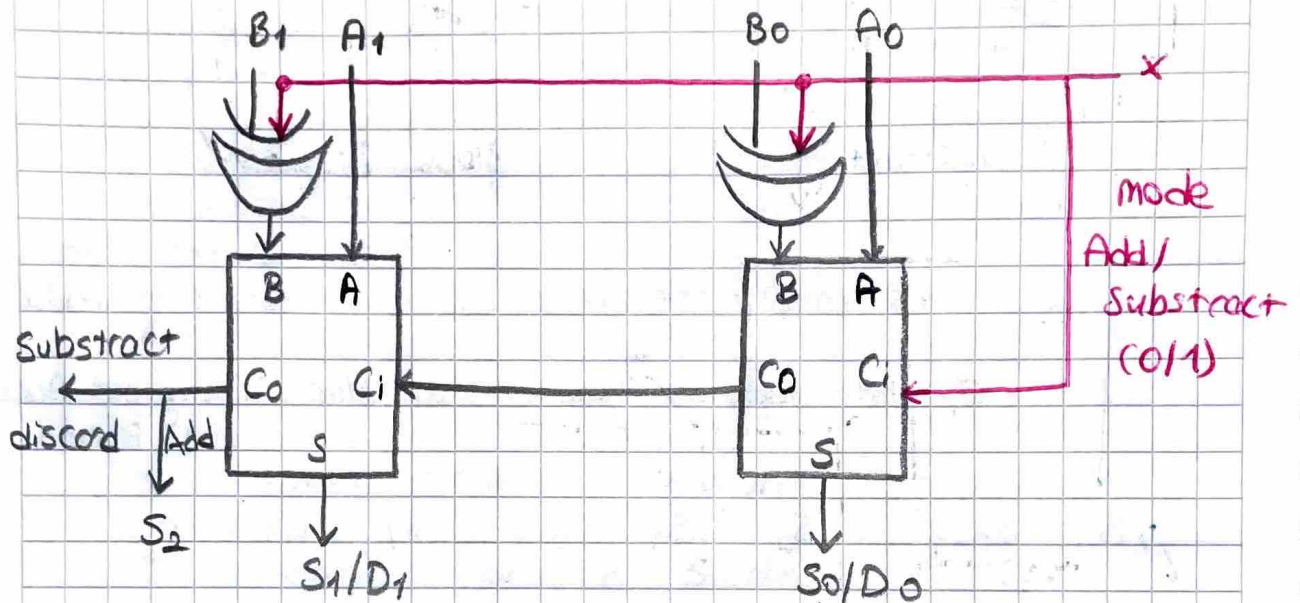
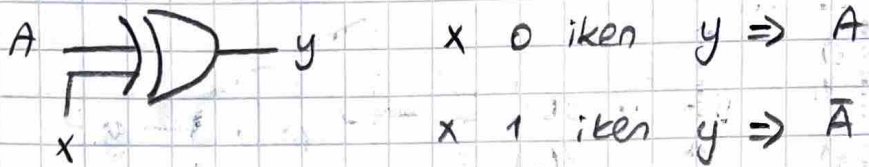
S_n S_{n-1} ... S_2 S_1 S_0 Sum / Difference

Add



→ EXOR kapısı aynen iletir (transfer modu 0 iken)

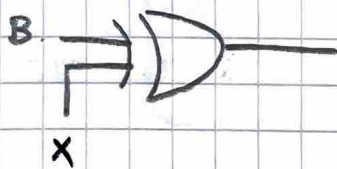
→ EXOR kapısı tünleyerek iletir (Evinin 1 modu)



Two Bits

2's Complement

Parallel adder / subtractor



$x = 0$ B transfer mode

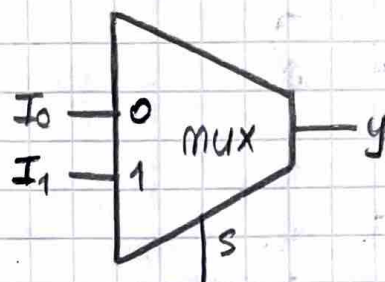
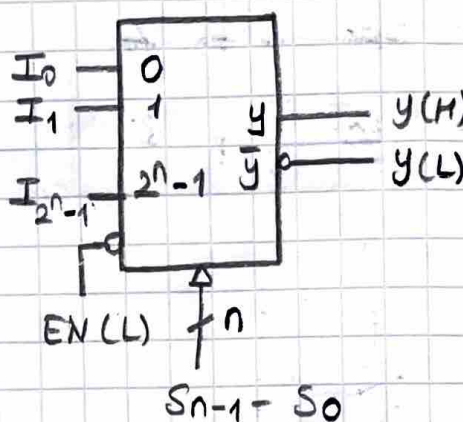
$x = 1$ $\bar{B}$ conversion mode

$$B \cdot \bar{x} + \bar{B} \cdot x$$

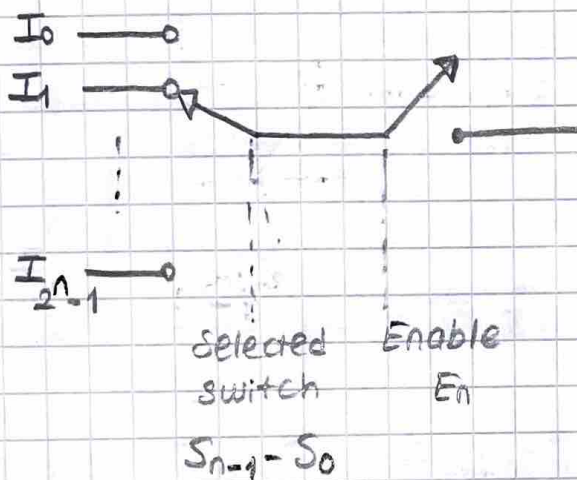
$$A \rightarrow A$$

$$-B \rightarrow \frac{t^n - B}{t^n - (B - A)}$$

MULTIPLEXER (MUX) / DATA SELEKTÖR



Block Diagram



MULTIPLEXER DESIGN

$$y_{\text{sop}} = \sum_{k=0}^{2^n-1} m_k \cdot I_k$$

min term

$$y_{\text{sop}} = (S_{n-1} \dots S_1 S_0) I_0 + (S_{n-1} \dots S_1 S_0) \cdot I_1 + \dots + (S_{n-1} \dots S_1 S_0) I_{2^n-1}$$

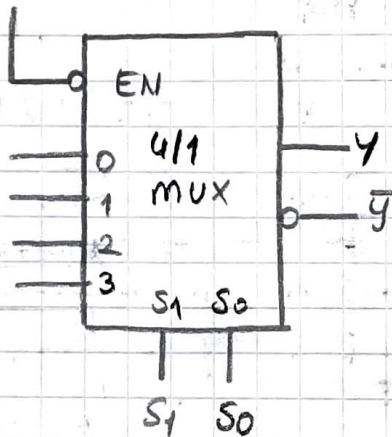
EN	S_{n-1}	...	S_0	y
0	x	...	x	0
1	0	...	0	I_0
1	0	...	1	I_1
⋮	⋮		⋮	⋮
1	1	1	1	I_{2^n-1}

4/1 MUX

$I_2 \dots I_1 I_0$	EN	S_1	S_0	y
0	X	X	X	0
1	0	0	0	I_0
1	0	1	1	I_1
1	1	0	0	I_2
1	1	1	1	I_3

S_1	S_0	0	1
0	$I_0 \overline{EN}$	$I_1 \overline{EN}$	
1	$I_2 \overline{EN}$	$I_3 \overline{EN}$	

/4



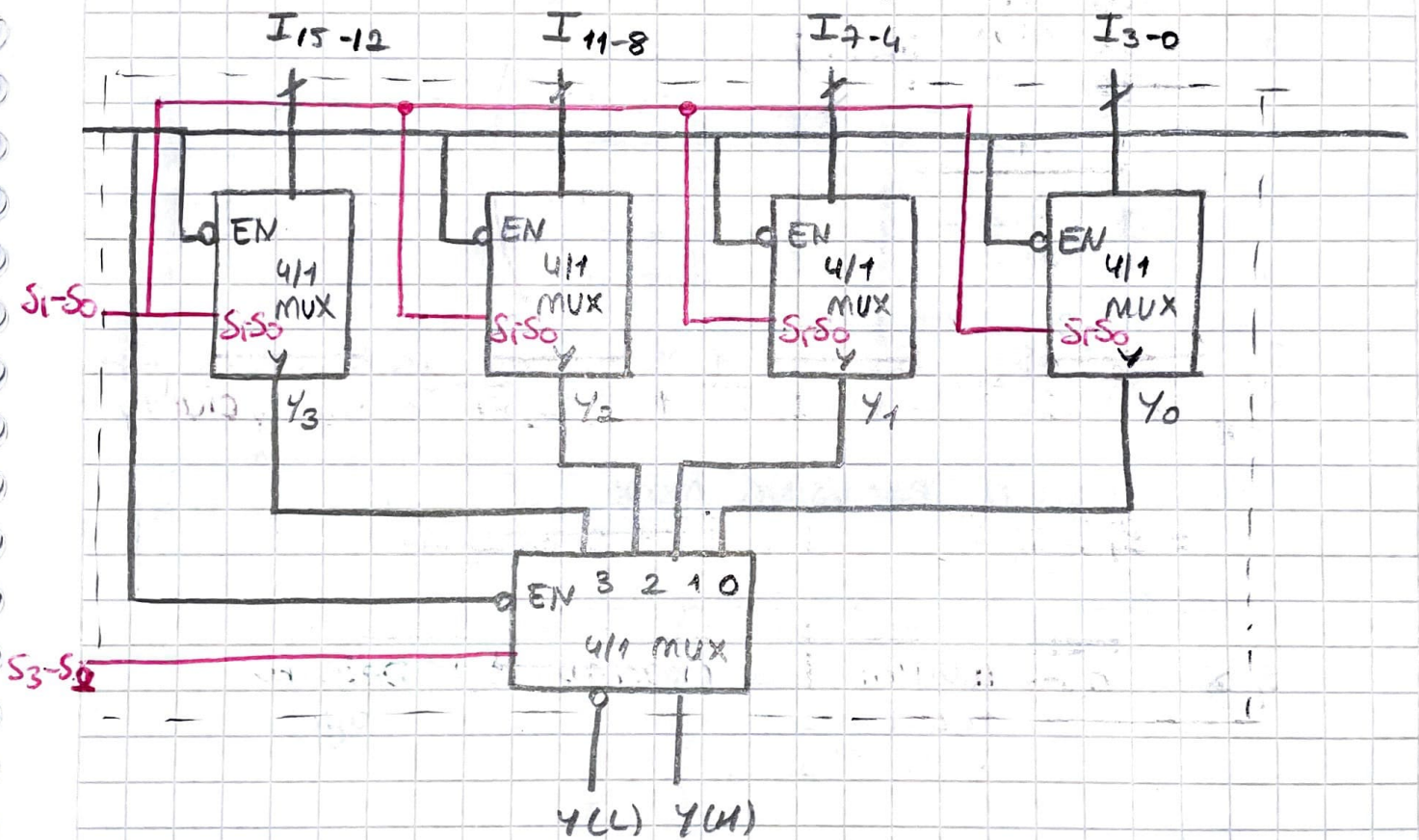
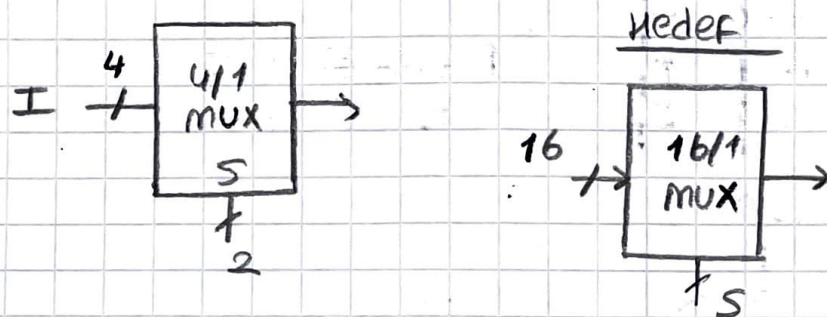
$$y_{sop} = m_0 I_0 \overline{EN} + m_1 I_1 \overline{EN} + m_2 I_2 \overline{EN} + m_3 I_3 \overline{EN}$$

$$= \overline{S_1} \overline{S_0} I_0 \overline{EN} + \overline{S_1} S_0 I_1 \overline{EN} + S_1 \overline{S_0} I_2 \overline{EN} + S_1 S_0 I_3 \overline{EN}$$

$$y_{sop} = \overline{\overline{y_{sop}}} = \overline{m_0 I_0 \overline{EN} \cdot m_1 I_1 \overline{EN} \cdot m_2 I_2 \overline{EN} \cdot m_3 I_3 \overline{EN}}$$

The Expansion Format for mux

ÖRN Elimizde 4/1 mux bulunsun.



16/1 mux

S_3	S_2	S_1	S_0	
0	0	0	0	I_0
0	0	0	1	I_1
0	0	1	0	I_2
0	0	1	1	I_3
0	1	0	0	I_4
0	1	0	1	I_5
0	1	1	0	I_6
0	1	1	1	I_7
1	0	0	0	I_8
1	0	0	1	I_9
1	0	1	0	I_{10}
1	0	1	1	I_{11}
1	1	0	0	I_{12}
1	1	0	1	I_{13}
1	1	1	0	I_{14}
1	1	1	1	I_{15}

$S_3 \ S_2 \ S_1 \ S_0 \rightarrow 1000$ seçtiğimiz sonuç I_8 olurdu

LOGIC DESIGN BY USING MUX

MUX

Logic Function

S | $n \rightarrow$ mux'un seçici giriş adedi | N (variable) $\rightarrow$ değişken sayısı

a) $n = N$

b) $n < N$

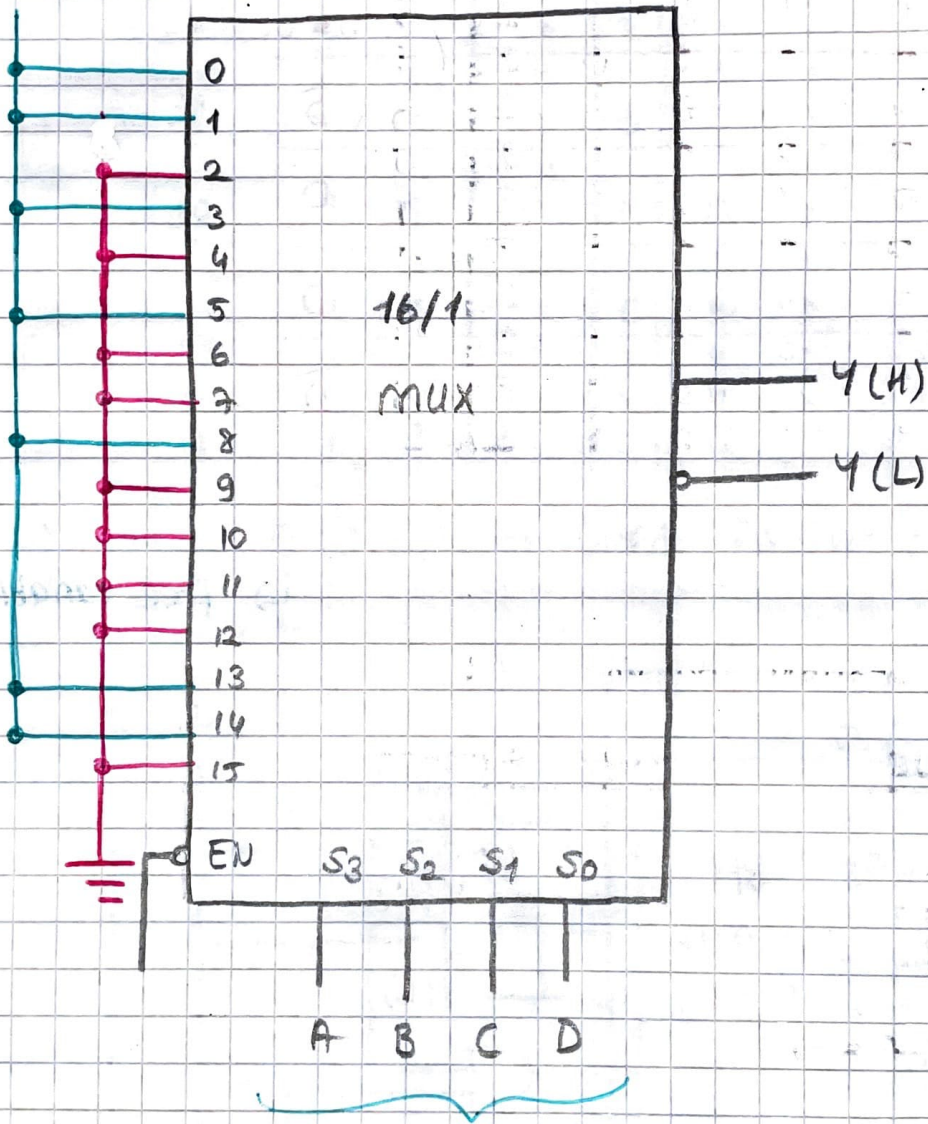
ÖRN

$$F = \sum m(0, 1, 3, 5, 8, 13, 14)$$

bu fonk.

- a) 16/1 mux kullanarak
- b) 8/1 mux kullanarak
- c) 4/1 mux kullanarak gerçekleştiriniz.

a) $n = 4$ $N = 4 \rightarrow n = N$

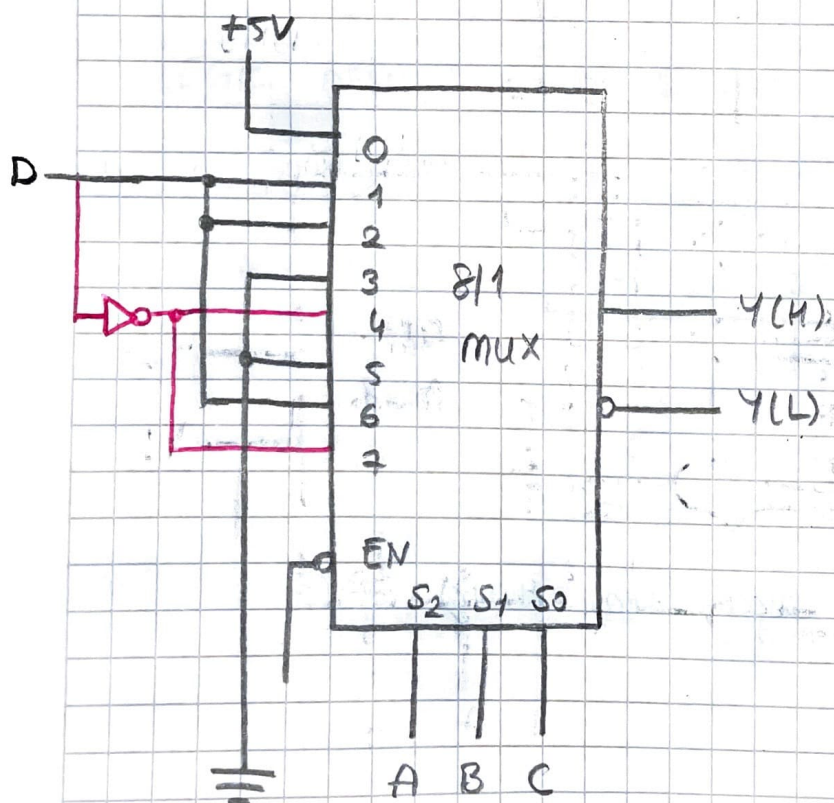


Dikkat, sırası önemli

	Dec	A	B	C	D	F	
I_0 0	0	0	0	0	0	1	1
	1	0	0	0	1	1	
I_1 1	2	0	0	1	0	0	
	3	0	0	1	1	1	D
I_2 2	4	0	1	0	0	0	
	5	0	1	0	1	1	D
I_3 3	6	0	1	1	0	0	
	7	0	1	1	1	0	0
I_4 4	8	1	0	0	0	1	
	9	1	0	0	1	0	$\bar{D}$
I_5 5	10	1	0	1	0	0	
	11	1	0	1	1	0	0
I_6 6	12	1	1	0	0	0	
	13	1	1	0	1	1	D
I_7 7	14	1	1	1	0	1	
	15	1	1	1	1	0	$\bar{D}$

b) $n=3$ $N=4$ $n < N \rightarrow A, B, C, D$

L) free variable



	Dec	A	B	C	D	F	
I_0 0	0	0	0	0	0	1	
	1	0	0	0	1	1	$\bar{C} + D$
	2	0	0	1	0	0	
	3	0	0	1	1	1	
I_1 1	4	0	1	0	0	0	
	5	0	1	0	1	1	$\bar{C} \cdot D$
	6	0	1	1	0	0	
	7	0	1	1	1	0	
I_2 2	8	1	0	0	0	1	
	9	1	0	0	1	0	$\bar{C} \cdot \bar{D}$
	10	1	0	1	0	0	
	11	1	0	1	1	0	
I_3 3	12	1	1	0	0	0	
	13	1	1	0	1	1	$C \oplus D$
	14	1	1	1	0	1	
	15	1	1	1	1	0	

c) $n=2$ $N=4$ $S_1 S_0$ ABCD
 $2^2 = 4$ $S_1 S_0$ free variable

